

Data Reduction

- **Data reduction:**

- Obtain a reduced representation of the data set
 - much smaller in volume but produces *almost* the same analytical results

- **Why data reduction?**—A database/data warehouse may store terabytes of data

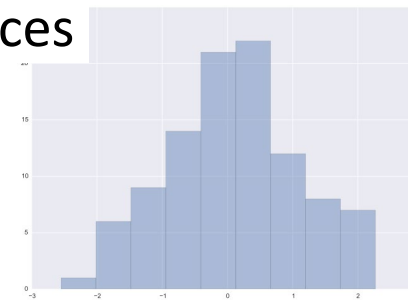
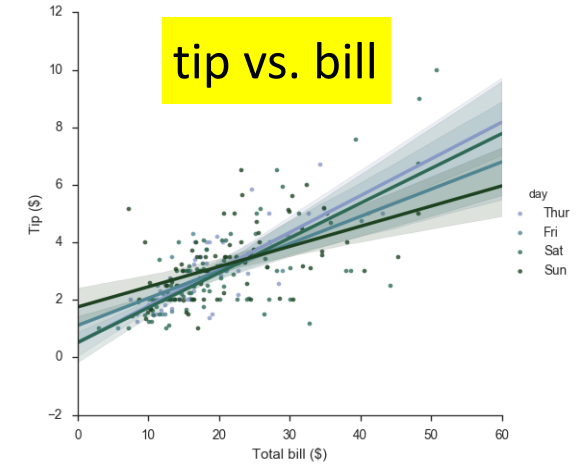
- Complex analysis may take a very long time to run on the complete data set

- **Methods for data reduction** (also *data size reduction* or *numerosity reduction*)

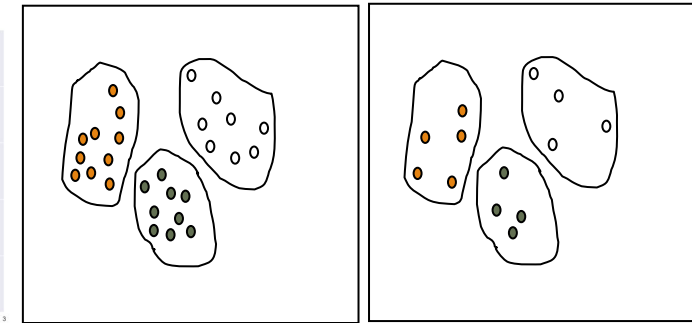
- Regression and Log-Linear Models
- Histograms, clustering, sampling
- Data cube aggregation
- Data compression

Data Reduction: Parametric vs. Non-Parametric Methods

- ❑ Reduce data volume by choosing alternative, *smaller forms* of data representation
- ❑ Parametric methods (e.g., regression)
 - ❑ Assume the data fits some model, estimate model parameters, store only the parameters, and discard the data (except possible outliers)
 - ❑ Ex.: Log-linear models—obtain value at a point in m -D space as the product on appropriate marginal subspaces
- ❑ Non-parametric methods
 - ❑ Do not assume models
 - ❑ Major families: histograms, clustering, sampling, ...



Histogram



Clustering on the Raw Data

Stratified Sampling